

Q.33 Differentiate between temperature switch and pressure switch.

Q.34 Explain butterfly valve.

Q.35 Draw and explain single loop temperature control system.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Explain PID controller with advantages and disadvantages.

Q.37 Explain principle and construction of diaphragm operated control valve.

Q.38 Explain process variables with example (Liquid level control system).

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5th Sem / IC, EI Sub : Process Control

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 What is the standard pressure range in I/P converter?

- a) 3-15 PSI b) 0-12 PSI
- c) 3-12 PSI d) 0-15 PSI

Q.2 In PID controller, "D" stands for :

- a) Delay b) Derivative
- c) Degree d) Dead Time

Q.3 Integral controller is also known as :

- a) Rate controller b) Reset controller
- c) Both d) None

Q.4 Flapper nozzle system is used in :

- a) Hydraulic system b) Electronic system
- c) Pneumatic system d) Mechanical system

Q.5 Offset problem mainly occurs in:

- a) P-Controller b) I-Controller
- c) D-Controller d) ON-OFF controller

Q.6 SI unit of pressure is:

- a) Pascal b) Newton
- c) Kelvin d) Candela

Q.7 PLC stands for:

- a) Programmable Logic Controller
- b) Proper Logic Control
- c) Process Level Controller
- d) None

Q.8 In automatic control system:

- a) Feedback is absent b) Feedback is present
- c) Sometimes present d) None

Q.9 Derivative controller is also called :

- a) Reset controller b) Rate controller
- c) Integral controller d) None

Q.10 Rotameter is fixed in:

- a) Horizontal Position b) Vertical position
- c) Any position d) Inclined position

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 Define process lag.

Q.12 Expand DCS.

Q.13 Define actuator.

Q.14 Write two merits of integral controller.

Q.15 What is measurement lag?

Q.16 Define switch.

Q.17 Write two process variables.

Q.18 Define dead zone.

Q.19 Expand PID.

Q.20 SI unit of force.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Explain ON-OFF control mode with diagram.

Q.22 Describe PI controller and its merits.

Q.23 Write short note on I/P converter.

Q.24 Explain flapper nozzle system.

Q.25 Differentiate between Pneumatic and Hydraulic systems.

Q.26 Describe proportional controller.

Q.27 Explain process and main process variables.

Q.28 Describe interlocking and sequencing circuit.

Q.29 Write short note on solenoid valve.

Q.30 Explain measurement lag and transmission lag.

Q.31 Write short note on auto transformer.

Q.32 Describe pneumatic actuator.